b) Implementing a Fuzzy Inference System

Aim:

To create a fuzzy logic system to evaluate temperature and output fan speed.

Procedure:

1. Define input/output variables.
2. Set membership functions.
3. Create rules and defuzzify output.

Python Code (using skfuzzy):

```
import numpy as np
import skfuzzy as fuzz
from skfuzzy import control as ctrl

# Input and Output variables
temperature = ctrl.Antecedent(np.arange(0, 41, 1), 'temperature')
fan_speed = ctrl.Consequent(np.arange(0, 101, 1), 'fan_speed')

# Membership functions
temperature['cold'] = fuzz.trimf(temperature.universe, [0, 0, 20])
temperature['warm'] = fuzz.trimf(temperature.universe, [10, 20, 30])
temperature['hot'] = fuzz.trimf(temperature.universe, [20, 40, 40])

fan_speed['low'] = fuzz.trimf(fan_speed.universe, [0, 0, 50])
fan_speed['medium'] = fuzz.trimf(fan_speed.universe, [25, 50, 75])
fan_speed['high'] = fuzz.trimf(fan_speed.universe, [50, 100, 100])

# Rules
rule1 = ctrl.Rule(temperature['cold'], fan_speed['low'])
rule2 = ctrl.Rule(temperature['warm'], fan_speed['medium'])
rule3 = ctrl.Rule(temperature['hot'], fan_speed['high'])

# Control system
fan_ctrl = ctrl.ControlSystem([rule1, rule2, rule3])
fan_sim = ctrl.ControlSystemSimulation(fan_ctrl)

# Input and compute
fan_sim.input['temperature'] = 30
fan_sim.compute()
print("Fan Speed:", fan_sim.output['fan_speed'])
```

Output:

Fan Speed: 80.55555555555556

Result:

Thus the Fuzzy Inference System for evaluating fan speed from temperature was implemented and executed successfully.